

Assignment 4: Automatic EBS Snapshot and Cleanup Using AWS Lambda and Boto3

Objective

The objective of this assignment is to automate the backup process for Amazon EBS volumes by creating snapshots using AWS Lambda and cleaning up old snapshots based on a defined retention period. For testing purposes, the retention period was set to 5 minutes.

AWS Services Used

- Amazon EC2 (EBS)
- AWS Lambda
- AWS IAM
- Amazon EventBridge (CloudWatch Events)
- Amazon CloudWatch Logs
- Boto3 (AWS SDK for Python)

Step-by-Step Implementation

Step 1: EBS Volume Setup

An existing EBS volume was identified from the EC2 dashboard. The volume ID was noted for use in the Lambda function to create snapshots.

vol-0d94ca1dc86a016a4

Last updated
1 minute ago

Actions

Del

Details			
Volume ID vol-0d94ca1dc86a016a4	Size 10 GiB	Type gp3	Status check Okay
AWS Compute Optimizer finding This user is not authorized to call AWS Compute Optimizer. Retry	Volume state Available	IOPS 3000	Throughput 125
Fast snapshot restored No	Availability Zone usw1-az1 (us-west-1a)	Created Thu Jan 01 2026 17:59:04 GMT+0530 (India Standard Time)	Multi-Attach enabled No
Attached resources -	Outposts ARN -	Managed false	Operator -

Source

Step 2: IAM Role Creation

An IAM role was created for the Lambda function with the following permissions:

- AmazonEC2FullAccess
- AWSLambdaBasicExecutionRole

This role allows the Lambda function to create and delete EBS snapshots and write logs.

Lambda-EBS-Snapshot-Role [Info](#)

Allows Lambda functions to call AWS services on your behalf.

Summary

Creation date

January 01, 2026, 18:01 (UTC+05:30)

Last activity

-

ARN

 arn:aws:iam::975050024946:role/Lambda-EBS-Snap

Maximum session duration

1 hour

Permissions[Trust relationships](#)[Tags](#)[Last Accessed](#)[Revoke sessions](#)

Permissions policies (2) [Info](#)

You can attach up to 10 managed policies.

[Simulate](#)[Remove](#)

Search		Filter by Type	
Search		All types	
☐	Policy name ↗	Type	Attached entities
☐	  AmazonEC2FullAccess	AWS managed	145
☐	  AWSLambdaBasicExecutionRole	AWS managed	95

Step 3: Lambda Function Creation

A Lambda function was created using Python 3.x runtime. The IAM role created earlier was assigned to the function.

[Lambda](#) > [Functions](#) > Lambda-EBS-Snapshot-Role

Successfully updated the function Lambda-EBS-Snapshot-Role.

Diagram | **Template**

 **Lambda-EBS-Snapshot-Role**

 Layers (0)

[+ Add trigger](#) [+ Add destination](#)

Description
-

Last modified
11 minutes ago

Function ARN
 arn:aws:lambda:us-west-1:975050024946:func:apshot-Role

Function URL | [Info](#)
-

Code | **Test** | **Monitor** | **Configuration** | **Aliases** | **Versions**

General configuration

Triggers

Permissions

Destinations

Function URL

General configuration [Info](#)

Description -	Memory 128 MB	Ephemeral storage 512 MB
Timeout 1 min 0 sec	SnapStart Info None	

Step 4: Lambda Code Logic

The Lambda function uses Boto3 to create a snapshot for the specified EBS volume. It then lists all snapshots associated with the volume and deletes snapshots older than 5 minutes.

Lambda > Functions > Lambda-EBS-Snapshot-Role

Successfully updated the function Lambda-EBS-Snapshot-Role.

Code source Info

```
1 import boto3
2 from datetime import datetime, timezone, timedelta
3
4 def lambda_handler(event, context):
5     ec2 = boto3.client('ec2')
6
7     VOLUME_ID = "vol-0d94ca1dc86a016a4"
8     RETENTION_MINUTES = 5 # For testing
9
10    # ----- CREATE SNAPSHOT -----
11    snapshot = ec2.create_snapshot(
12        VolumeId=VOLUME_ID,
13        Description="Automated snapshot created by Lambda"
14    )
15
16    snapshot_id = snapshot['SnapshotId']
17    print(f"Created snapshot: {snapshot_id}")
18
19    # ----- CLEANUP OLD SNAPSHOTS -----
```

PROBLEMS OUTPUT CODE REFERENCE LOG TERMINAL

Status: Succeeded
Test Event Name: hello-world

Response:

```
{
  "statusCode": 200,
  "body": "EBS snapshot creation and cleanup completed"
}
```

The area below shows the last 4 KB of the execution log.

Step 5: Testing and Verification

The Lambda function was manually invoked. A new snapshot was created successfully. After waiting for more than 5 minutes, the function was triggered again and old snapshots were deleted as expected.

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search

Clear 1m 30m 1h 12h Custom UTC timezone Display

Timestamp	Message
	No older events at this moment. Retry
2026-01-01T12:37:57.422Z	INIT_START Runtime Version: python:3.14.v32 Runtime Version ARN: arn:aws:lambda:us-west-1::runtime:1ee4e6d61a50fbb29d03b87572cc627d0a92de84538be7b218...
2026-01-01T12:37:57.735Z	START RequestId: 751f0590-2bd1-42ae-81a2-1730d6be9088 Version: \$LATEST
2026-01-01T12:38:01.282Z	Created snapshot: snap-0e033e784a371302b
2026-01-01T12:38:01.339Z	No snapshots older than 5 minutes found
2026-01-01T12:38:01.371Z	END RequestId: 751f0590-2bd1-42ae-81a2-1730d6be9088
2026-01-01T12:38:01.371Z	REPORT RequestId: 751f0590-2bd1-42ae-81a2-1730d6be9088 Duration: 3635.04 ms Billed Duration: 3945 ms Memory Size: 128 MB Max Memory Used: 100 MB Init...
	No newer events at this moment. Auto retry paused. Resume

Result

The Lambda function successfully automated EBS snapshot creation and cleanup. Snapshots older than the defined retention period were deleted, ensuring backup automation and cost optimization.

GitHub Repository

GitHub Link: <https://github.com/Rushiargade/aws-lambda-ebs-snapshot-cleanup.git>